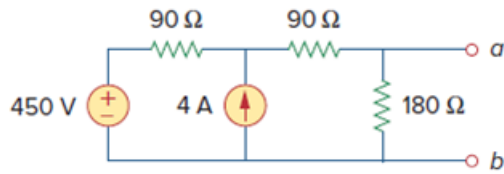


Problem

Find the Norton equivalent circuit for the circuit in Fig. 4.42, at terminals a - b .

Figure. 4.42:



Step-by-step solution

Step 1 of 5

Refer to the circuit diagram in Figure 4.42 in the textbook.

Determine the Norton equivalent resistance.

Short circuit the independent voltage sources and open circuit the current sources as shown in Figure 1.

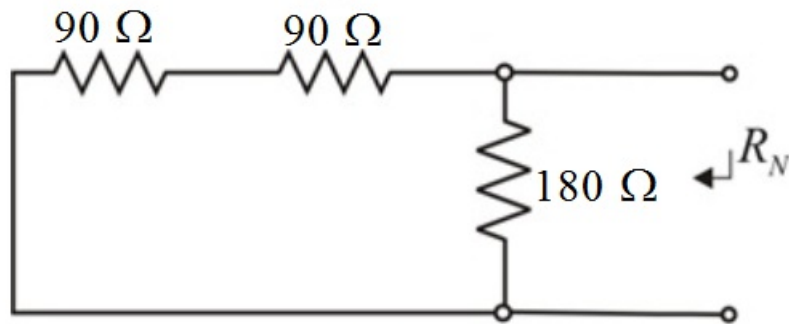


Figure 1

Resistors $90\ \Omega$ and $90\ \Omega$ are in series and the series combination is in parallel with the resistor $180\ \Omega$.

Determine Norton resistance R_N .

$$\begin{aligned} R_N &= (90 + 90) \parallel (180) \\ &= 180 \parallel 180 \\ &= 90\ \Omega \end{aligned}$$

Therefore, the Norton resistance R_N is $90\ \Omega$.

Step 2 of 5

Short circuit the terminals a and b as shown in Figure 2.

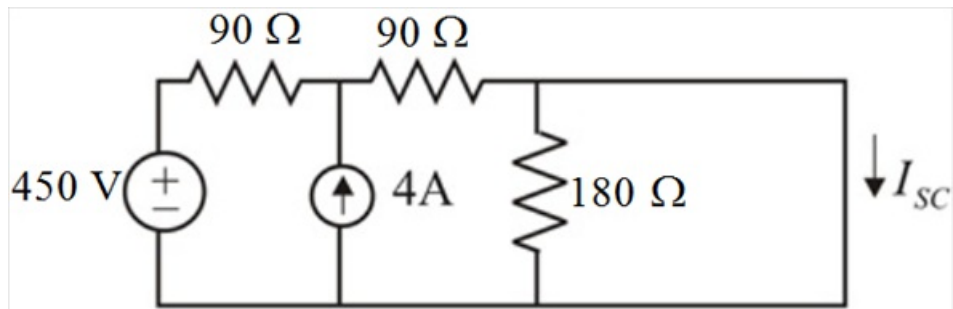


Figure 2

Ignore the $180\ \Omega$ resistor because it has been short circuited.

Redraw the circuit diagram as shown in figure 3.

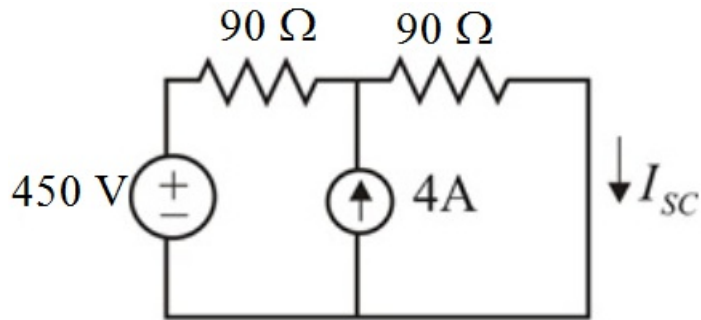


Figure 3

Step 3 of 5

Apply source transformation technique to convert the 450 V and $90\ \Omega$ into current source in parallel with the resistor.

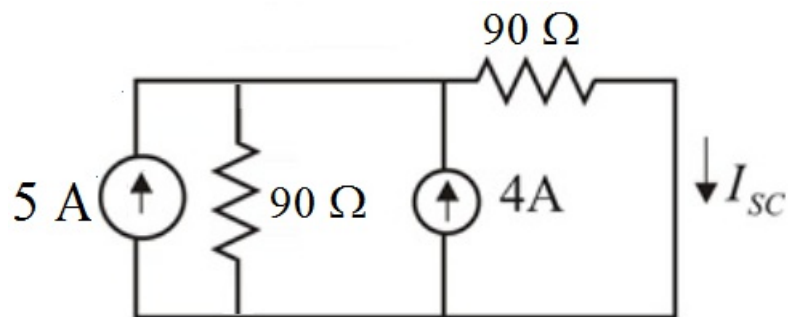


Figure 4

Add the current sources as they are parallel to each other.

Redraw the circuit as shown in Figure 5.

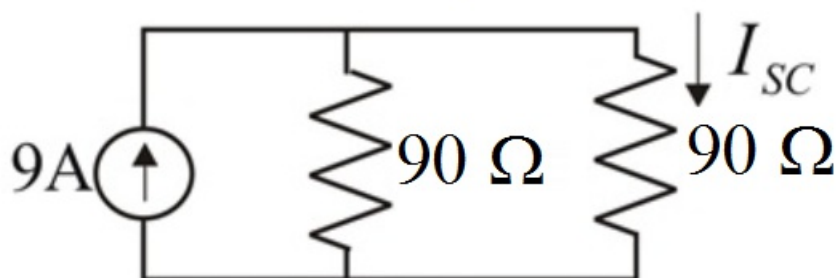


Figure 5

Step 4 of 5

Determine current I_{sc} .

Apply current division principle to the circuit shown in Figure 5.

$$\begin{aligned} I_{sc} &= 9 \times \frac{90}{90+90} \\ &= 9 \times \frac{90}{180} \\ &= 4.5 \text{ A} \end{aligned}$$

Therefore, Norton current I_{sc} is 4.5 A.

Step 5 of 5

Draw the Norton equivalent circuit as shown in Figure 6.

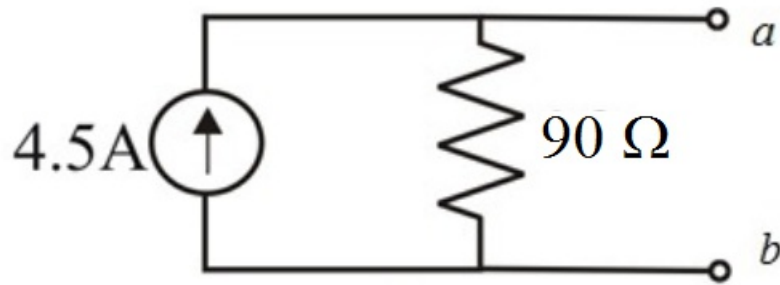


Figure 6

Thus, Norton equivalent circuit has been measured.